

Navigation Strategies & Algorithms

Open Challenge Strategy

- **Lane Centering:** The control algorithm maintains the vehicle centered between the inner and outer track walls, with a slight algorithmic bias prioritizing proximity to the inner boundary wall to optimize cornering radii.
- **Lap Counter & Stop Condition:** Track lap progression is monitored by counting corner marker lines via the camera. Once **11 corner lines** are detected, the system transitions state: camera navigation yields to **encoder odometry**, driving the vehicle a pre-calculated physical distance before bringing it to a full stop.

Obstacle Challenge Strategy

- **Centering & Alignment:** The camera module is fixed along the chassis centerline. The pixel distance between a detected pillar's centroid and the frame centerline represents the relative angular offset of the obstacle ($\theta = 0^\circ$ at true center).
- **Dynamic Steering Angle Reduction:** The target avoidance angle between the chassis and the pillar decreases linearly as the pillar's pixel footprint increases in the camera frame (indicating closer proximity). This creates a smooth, sweeping curve around pillars rather than abrupt, late swerves.